Final Project Report Rubric & Guidelines

For the final project report, you will submit a complete write-up of your work from the semester. This report should build upon your project proposal and present your completed research, experiments, results, and analysis.

Rubric for Track 1

- Page Limit: No more than 6 pages (references do not count toward the page limit).
- PDF Format: NeurIPS template (https://www.overleaf.com/latex/templates/neurips-2024/tpsbbrdqcmsh)
- Submission: Report (PDF) + Code (with a README explaining how to reproduce results)
- You may work individually or in teams of up to 3 members. **Each team only needs to submit one report.**

1. Introduction & Problem Definition (10 pts)

Provide a concise introduction to the problem you investigated.

- Problem Statement: Clearly define the specific problem you addressed.
- Motivation & Significance: Why is this problem interesting or socially/technically important?
- Learning Objective & Methodology: What exactly is the model tasked to "learn" (e.g., node labels, edge existence, graph-level properties)? What method is applied to tackle the problem?
- Summary: A conclusion on the experimental results produced by the models.

2. Dataset (10 pts)

Describe the dataset(s) used in your project:

- Description: Source of the data and its real-world meaning.
- Exploratory Data Analysis (EDA):
  - Graph Statistics: Number of nodes/edges, density, average clustering coefficient, and degree distribution.

- Feature Analysis: Brief description of node/edge attributes and label distribution (note any class imbalance).
- Visualizations: Provide at least one meaningful visualization (e.g., a sample subgraph, t-SNE plot of learned embeddings, or degree distribution histogram).
- Data Splits: Describe how the data was split for training, validation, and testing.

3. Related Works (10 pts)

Provide a literature review that contextualizes your work:

- Breadth & Depth: Cite at least 3–5 high-quality papers relevant to the domain or methodology.
- Comparative Analysis: Do not just list papers. Explain how these works succeed, where they fall short, and how your project differs or extends their ideas.
- Positioning: Clearly state how your work relates to and builds upon prior research (e.g., "Unlike [Author et al.], our project focuses on...").

4. Methodology (15 pts)

Describe the approaches you implemented in detail. This section should reflect what you actually built, not just what you proposed.

- Individual Contribution: Each team member must have a dedicated subsection (e.g., Section 4.1, 4.2). Each member's approach should contain unique elements.
- Technical Details: Provide sufficient detail for someone to reimplement your method: architecture diagrams, key equations, layer configurations, etc.
- Technical Justification: Why were these specific architectures or techniques chosen for your dataset and problem?
- Adaptation/Originality: If using existing code, clearly specify your non-trivial modifications (e.g., custom loss function, novel attention mechanism, feature engineering pipeline). Clearly distinguish between borrowed code and your own contributions.

5. Experiments & Results (35 pts)

This is the most important section of the report. Present a thorough and rigorous experimental evaluation.

- Experimental Setup (5 pts):
  - Baselines: List the standard models you compared against (e.g., MLP, vanilla GCN, heuristic methods).
  - Hyperparameters: Report key hyperparameters and how they were selected (grid search, random search, etc.).

  - Computational details: Hardware used, training time, and any relevant implementation details.

- Quantitative Results (15 pts):
  - Present results in well-organized tables and/or figures.
  - Use appropriate metrics (Accuracy, F1-score, AUROC, RMSE, etc.) and justify why these metrics are suitable for your problem.
  - Report results with standard deviations over multiple runs where applicable.
  - Compare all proposed methods against baselines clearly.

- Analysis & Discussion (15 pts):
  - What worked and what didn't? Interpret them.
  - Ablation Studies: If applicable, include ablation experiments to isolate the contribution of individual components (e.g., removing an attention mechanism, changing the number of layers).
  - Qualitative Analysis: Where possible, provide qualitative examples (e.g., visualizations of learned embeddings, attention weights, or case studies of correct/incorrect predictions).
  - Error Analysis: Identify failure modes or systematic weaknesses of your approach.
  - Answering Research Questions: Revisit the specific research questions and hypotheses from your proposal. Were they confirmed or refuted?

6. Conclusion (10 pts)

- Summary: Briefly summarize the key findings and contributions of your project.
- Lessons Learned: What were the main challenges you encountered and how did you address them?
- Limitations: Acknowledge the limitations of your approach.

7. Code Quality & Reproducibility (10 pts)

- Code Organization: Is the code well-structured and documented?
- README: Does the repository include a clear README with instructions for reproducing results (dependencies, data download, training commands)?
- Reproducibility: Can your results be reproduced by running the provided code?

Rubric for Track 2

- Page Limit: No more than 6 pages (references do not count toward the page limit).
- PDF Format: NeurIPS template (https://www.overleaf.com/latex/templates/neurips-2024/tpsbbrdqcmsh)
- Submission: Report (PDF) + Code (with a README explaining how to reproduce

results)
- Baseline Requirement: You must outperform the provided baselines
- You must work individually

1. Dataset (10 pts)

Provide a detailed description of the competition dataset:

- Exploratory Data Analysis (EDA):
  - Graph Statistics: Number of nodes/edges, density, average clustering coefficient, and degree distribution.
  - Feature Analysis: Brief description of node/edge attributes and label distribution (note any class imbalance).
- Visualizations: Provide at least one meaningful visualization (e.g., a sample subgraph, t-SNE plot of features, or degree distribution).
- Key Observations: Highlight any dataset characteristics that informed your design decisions (e.g., high heterophily, skewed degree distribution, sparse features).

2. Literature Review & Comparative Analysis (10 pts)

- Breadth & Depth: Cite at least 3–5 high-quality academic papers relevant to the architecture or the specific graph task you implemented.
- Positioning: Explain how your approach relates to these works (e.g., "Inspired by [Author et al.], we adopted a multi-head attention mechanism but modified the aggregation function to better handle skewed degree distributions.").

3. Methodology & Technical Implementation (35 pts)

Describe what you actually implemented, with full technical detail. Your methodological novelty and contributions should come from one or more of the following areas:

- Data-Side Novelties (Graph & Feature Engineering):
  - Graph Rewiring/Augmentation: How did you modify the graph topology? (e.g., virtual nodes, edge dropping/adding, spatial/spectral sparsification).
  - Feature Augmentation: Powerful node/edge features (e.g., Positional/Structural Encodings like Laplacian eigenvectors, Random Walk features, graph motifs).
  - Sampling Strategies: Custom neighbor sampling or handling class imbalance (e.g., SMOTE for graphs).

- Model-Side Novelties (Architecture Design):
  - Custom Aggregation/Message Passing: Modified information aggregation (e.g., custom attention, heterophily-aware designs).
  - Architectural Enhancements: Advanced components like Jumping Knowledge

networks, GNN + Graph Transformer hybrids, residual connections, and normalization strategies.

- Pipeline-Side Novelties (Training & Inference):
  - Custom Loss Functions: Loss functions tailored to the task (e.g., Focal Loss, Contrastive Loss, margin-based losses).
  - Training Dynamics: Advanced strategies (e.g., curriculum learning, adversarial training, knowledge distillation).
  - Post-processing & Ensembling: Label propagation (e.g., Correct & Smooth), ensemble methods combining diverse GNN models.

- Technical Justification: Clearly explain why each design choice was made for this specific dataset and task.

4. Experiments & Results (35 pts)

Present a thorough experimental evaluation of your solution.

- Experimental Setup (5 pts):
  - Baselines: List all models you compared against, including the provided competition baselines.
  - Hyperparameters: Report key hyperparameters and selection strategy.
  - Computational details: Hardware, training time, and implementation specifics.

- Quantitative Results (15 pts):
  - Present results in well-organized tables and/or figures.
  - Compare against the provided baselines and any additional baselines you implemented.
  - Report standard deviations over multiple runs where applicable.

- Analysis & Discussion (15 pts):
  - What worked and what didn't? Interpret your results. Don't only report numbers.
  - Ablation Studies: Isolate the contribution of individual components. For example, if you used both feature augmentation and a custom loss, show results with and without each.
  - Qualitative Analysis: Visualizations of learned embeddings, attention weights, or example predictions where possible.
  - Error Analysis: Identify failure modes or patterns in incorrect predictions.

5. Conclusion (5 pts)

- Summary: Key findings and what contributed most to your performance.
- Limitations: Acknowledge weaknesses or areas where performance could improve.

6. Code Quality & Reproducibility (5 pts)

- Code Organization: Is the code well-structured and documented?
- README: Clear instructions for reproducing your leaderboard results.
- Reproducibility: Can your final submission score be reproduced?